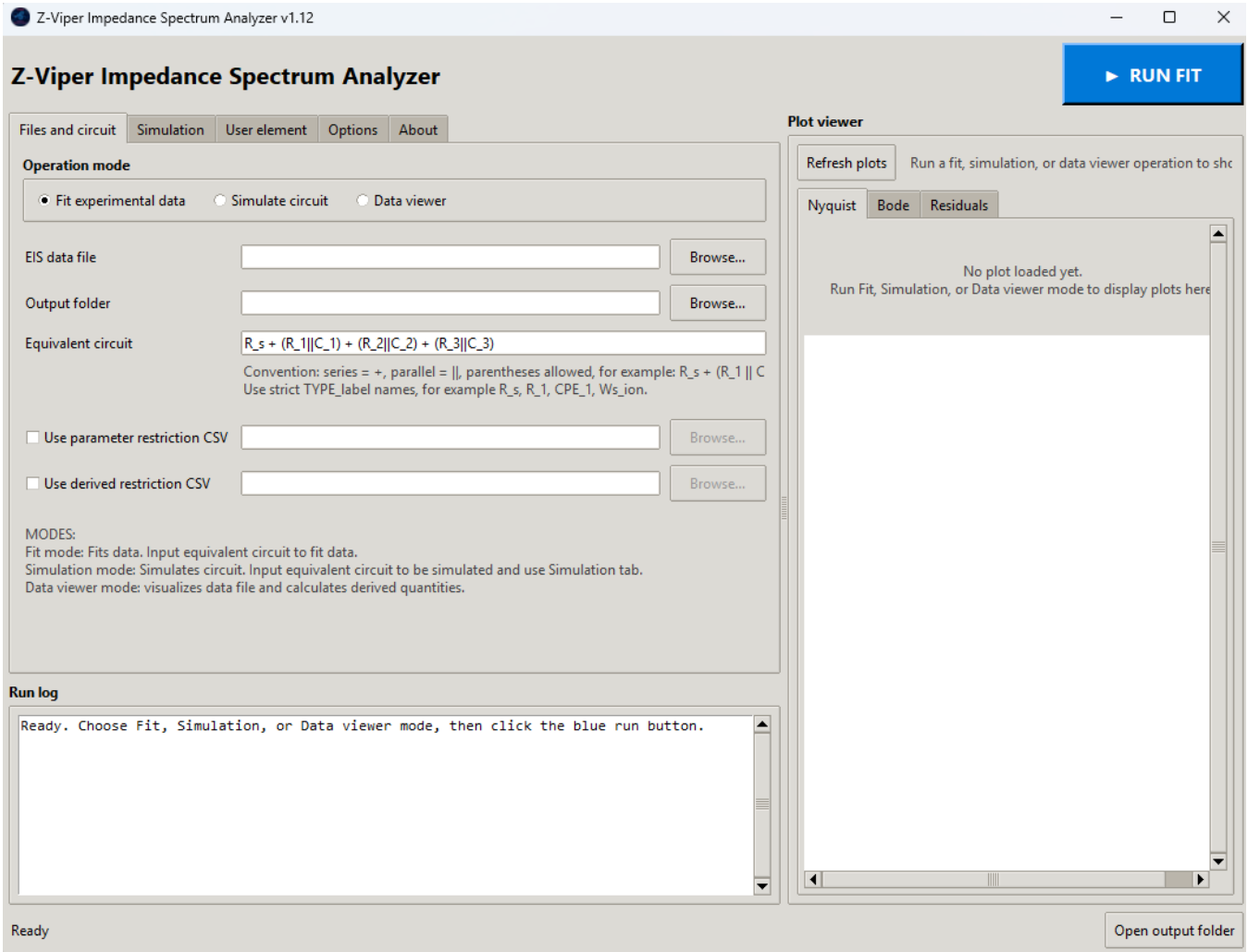
# Z-Viper
# Impedance Spectrum Analyzer
## v1.12
## User Manual





# Contents

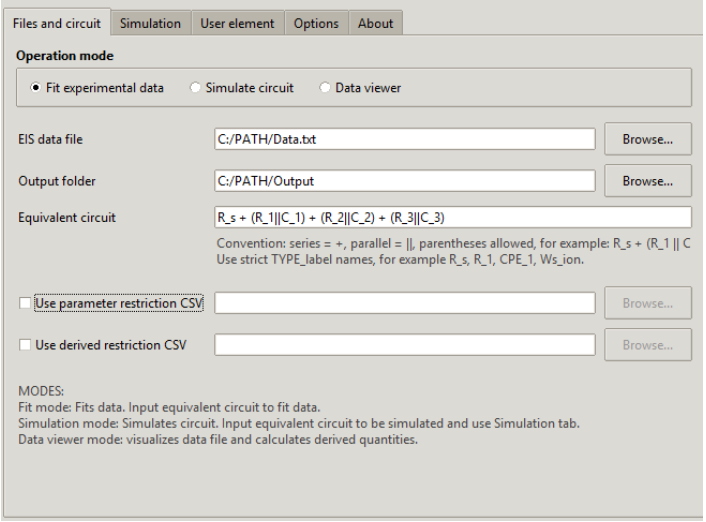
# 1. Installation and launch

**Using the .exe file:**
Download the Z-Viper_Impedance_Spectrum_Analyser_v1.12.exe file.
Run the `.exe` file.

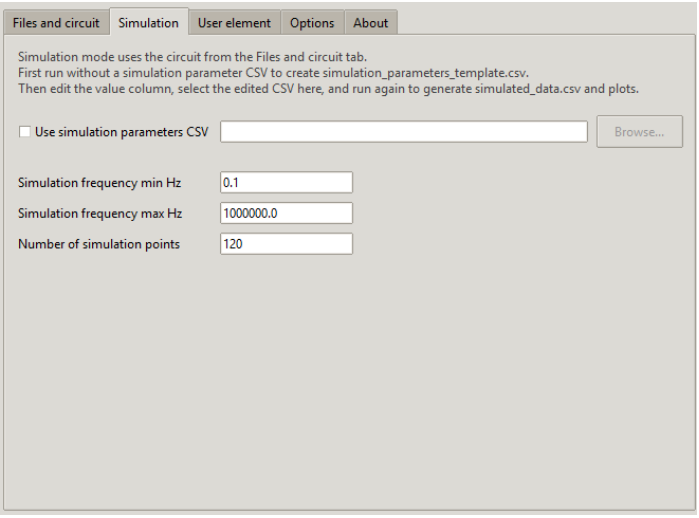
# 2. GUI structure

The software is divided into different tabs:

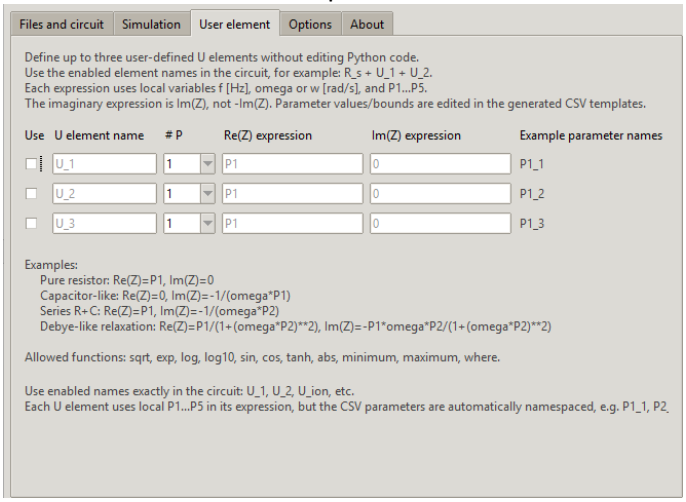
- **Files and circuit**: choice of the operation mode (see Section 3), path to data file and output folder, input of the equivalent circuit (see Section 4), and input of optional fit restriction files.



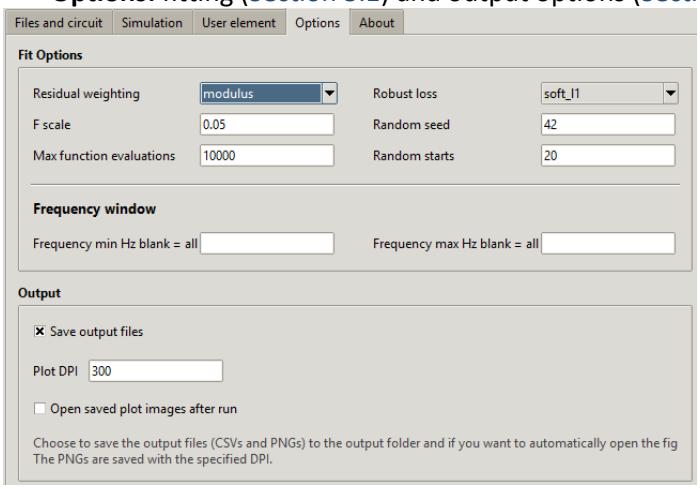

- **Simulation**: to be used when running the Simulation mode (see Section 3.2)



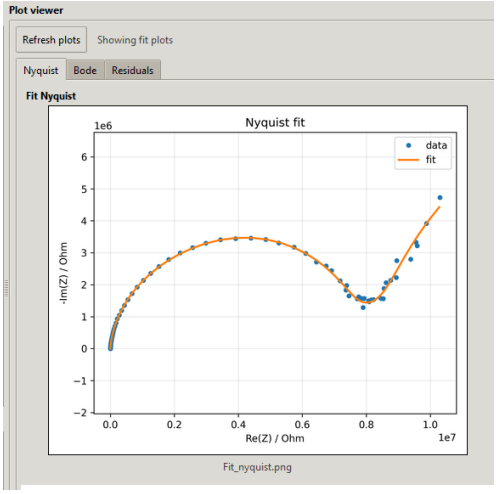
- **User element**: define up to three different user-defined circuit elements (see Section 5).

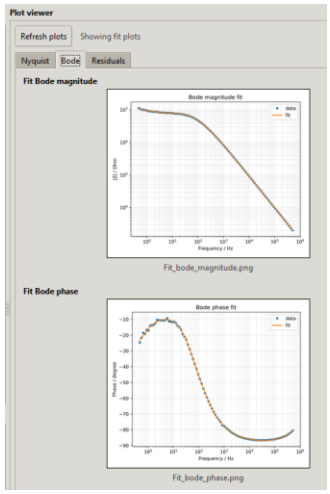


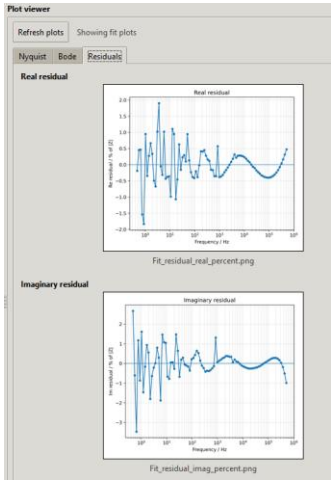

- **Options:** fitting (Section 3.1) and output options (Section 6).



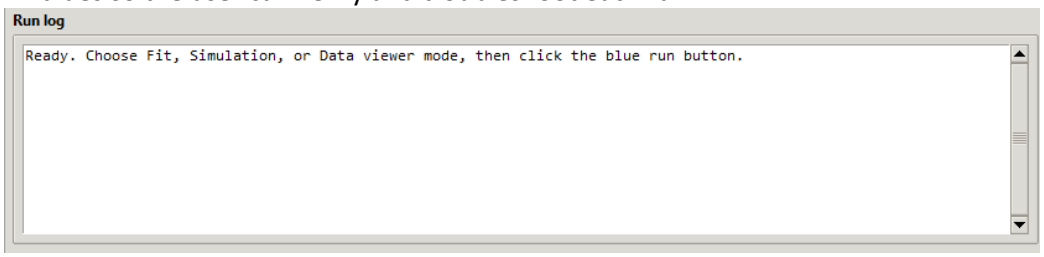

- **Plot viewer:** The Plot viewer displays the generated figures directly inside the GUI after a successful run. It is organized into separate tabs for Nyquist, Bode, and residual plots, allowing the user to quickly visually inspect the output. The `Refresh plots button` reloads the figures from the selected output folder, which is useful if plots were regenerated or updated during a new run.



- **Run Log**: displays real-time progress, warnings, errors, output-file messages, fit metrics, and final parameter values so the user can verify and troubleshoot each run.




# 3. Operation modes

## 4.1 Fit experimental data

Fits an experimental impedance spectrum with the selected equivalent circuit, extracting best-fit element values, residuals, fit metrics, parameter tables, and Nyquist/Bode/residual plots.

### Input

- `Data file` (.txt) The standard experimental input format is:

```
line 1: number of points
line 2 onward: Re(Z), -Im(Z), Frequency
```

The program assumes the second column is -Im(Z), which is positive for normal capacitive arcs.

- `Output folder`: Folder where all the output files will be saved
- `Circuit string`: String defining the equivalent circuit. See Section 4.

### Output

| File | Description |
|---|---|
| `Fit_data.csv` | Contains the measured data, fitted impedance values, impedance magnitude/phase, and real/imaginary residuals at each frequency point. |
| `Fit_parameters.csv` | Contains the final fitted parameter values together with their initial values, bounds, vary/fixed status, units, and descriptions. |
| `Fit_parameter_restrictions_template.csv` | Template file listing fit parameters, starting values, hard bounds, vary/fixed status, units, and descriptions. *This file is generated only if the "Use parameter restriction CSV" option is NOT selected.* |
| `Fit_derived_restrictions_template.csv` | Template file for optional derived physical constraints, such as branch peak frequencies, Warburg transition frequencies, and parameter ratios. *This file is generated only if the "Use derived restriction CSV" option is NOT selected.* |
| `Fit_nyquist.png` | Nyquist plot showing the experimental impedance data together with the fitted circuit response. |
| `Fit_bode_magnitude.png` | Bode magnitude plot showing experimental and fitted \|Z\| as a function of frequency. |
| `Fit_bode_phase.png` | Bode phase plot showing experimental and fitted phase angle as a function of frequency. |
| `Fit_residual_imag_percent.png` | Plot of the imaginary-part residual as a percentage of measured \|Z\| versus frequency. |
| `Fit_residual_real_percent.png` | Plot of the real-part residual as a percentage of measured \|Z\| versus frequency. |
| `Fit_model_summary.txt` | Summary of the fitted circuit, data file, active settings, optimizer result, fit metrics, and best-fit parameters. |
| `Fit_optimizer_history.csv` | Records the outcome of each optimizer start, including cost, success flag, number of function evaluations, and optimizer message. |
| `Fit_metrics.csv` | Summarizes the goodness-of-fit values, including RMSE, relative RMSE, chi-square, AIC, and BIC. |

### Fitting procedure

1. Select **Fit experimental data** (Files and circuit tab).
2. Select the data file.
3. Select an output folder.
4. Enter the equivalent circuit (see Section 4).
5. Optionally select a parameter restriction CSV.
6. Optionally select a derived restriction CSV.
7. Choose settings in Fit options.
8. Click RUN FIT.

During fitting, the program writes templates, builds default parameter guesses, applies restrictions, runs nonlinear least-squares from multiple starts, and saves CSV outputs and plots.

### Parameter restriction CSV

*The program writes parameter_restrictions_template.csv if no file is selected*. Edit it and select it as the parameter restriction CSV for later fits.

| Column | Meaning |
|---|---|
| parameter | Parameter name, optionally with unit in brackets. |
| value | Initial value used by the optimizer. |
| min | Hard lower bound. |
| max | Hard upper bound. |
| vary | True = fit this parameter; False = fix this parameter. |
| description | Human-readable explanation. |

Example:

```
parameter,value,min,max,vary,description
R_s [Ohm],250,100,1000,True,Series/contact resistance
R_1 [Ohm],1.0E6,1.0E4,1.0E8,True,Resistance
Q_1 [S*s^n],1.0E-9,1.0E-12,1.0E-6,True,CPE Q parameter
n_1 [-],0.90,0.50,1.00,True,CPE exponent
```

The value column is the starting guess, not the final value. If vary is False, the parameter is fixed at value.

**Important note**: More complex equivalent circuits can have multiple “minima” when fitted, leading to good fits but with unphysical values for the parameters. Thus, it might be important to restrict the parameters to physically meaningful values.

### Derived/physical restriction CSV

*The program writes derived_restrictions_template.csv if no file is selected*. Derived restrictions are soft penalties applied to calculated physical quantities rather than direct parameters.

| Type | Meaning |
|---|---|
| RCPE_PEAK_HZ | Peak/characteristic frequency of an R\|\|CPE branch. |
| RC_PEAK_HZ | Peak frequency of an R\|\|C branch. |
| RL_TRANSITION_HZ | Transition frequency of an R\|\|L branch. |
| RLPE_TRANSITION_HZ | Transition estimate for an R\|\|LPE branch. |
| L_TRANSITION_HZ | Transition estimate for an L\|\|LPE branch. |
| WS_TRANSITION_HZ | Approximate transition frequency of a short Warburg. |
| WS_LF_RESISTANCE_OHM | Low-frequency resistance scale of a short Warburg. |
| WO_TRANSITION_HZ | Approximate transition frequency of an open Warburg. |
| WO_LF_RESISTANCE_OHM | Low-frequency resistance scale of an open Warburg. |
| PARAMETER_RATIO | Ratio between two parameters. |

Important columns are enabled, name, type, parameter-name columns such as R/Q/n/Wsr/Wsc, min, max, weight, and note. Disabled rows are ignored.

**Important note**: This restriction constrains physical quantities calculated from fitted parameters, not the parameters themselves. This is particularly useful to avoid “branch swapping” which avoids confusion when comparing the same device under different conditions. This is optional and to be used only at later stages of the fit.

## Fit options and optimization

The fit options are found in the `Options` tab.

### Residual weighting

| Option | Meaning | Recommended use |
|---|---|---|
| `modulus` | Residual divided by \|Z\| point-by-point. | Recommended default for EIS. |
| `sqrt_modulus` | Residual divided by sqrt(\|Z\|). | Intermediate weighting. |
| `none` | Unweighted residual in ohms. | Use only when absolute-ohm errors should dominate. |

### Robust loss

| Loss | Behavior | Recommendation |
|---|---|---|
| `linear` | Ordinary least squares; large residuals dominate strongly. | Strict comparison only. |
| `soft_l1` | Smooth robust loss; large residuals are downweighted gradually. | Best default. |
| `huber` | Least squares for small residuals, linear penalty for large residuals. | Good robustness cross-check. |
| `cauchy` | Strongly suppresses large residuals. | Diagnostic only. |
| `arctan` | Very aggressive limiting of very large residuals. | Stress test only. |

### F Scale

F_SCALE controls where robust loss starts treating residuals as large. With modulus weighting, 0.05 roughly corresponds to a 5% normalized residual transition. Larger values are closer to ordinary least squares; smaller values are more robust. F_SCALE is ignored when LOSS = linear.

| F scale | Effect |
|---|---|
| `0.10` | Closer to ordinary least squares; less robust. |
| `0.05` | Good default. |
| `0.03` | More robust; downweights large residuals earlier. |
| `0.01` | Very aggressive; can hide real low-frequency behavior. |

### Random starts and random seed

N_RANDOM_STARTS controls how many randomized initial guesses are tried in addition to the main starting guess. RANDOM_SEED makes those random starting guesses reproducible. If different seeds give nearly the same result, the solution is stable with respect to initialization. If different seeds give significantly different results, the model is non-unique or the bounds need improvement.

## Recommended defaults

```
WEIGHT_MODE = modulus
LOSS = soft_l1
F_SCALE = 0.05
N_RANDOM_STARTS = 20 to 50
RANDOM_SEED = 42
MAX_NFEV = 10000 to 30000
DPI = 300
```

For robustness checking, use LOSS = huber with F_SCALE = 0.05. For a more least-squares-like check, use LOSS = soft_l1 with F_SCALE = 0.10. Use cauchy only as an outlier diagnostic.

## 4.2 Simulation

Generates an impedance spectrum from the selected equivalent circuit using values supplied in a simulation-parameter CSV file.

**Input:**

- `Simulation parameters` (.csv). This file gives the parameters for the circuit elements. If no file is input, the program will generate a template the user can modify.
- `Circuit string`: String defining the equivalent circuit. See section X
- `Output folder`: Folder where all the output files will be saved

**Output:**

| File | Description |
|---|---|
| `Simulation_parameters_template.csv` | Template file listing all circuit parameters needed for simulation; edit the value column and use it as the simulation-parameter input file. *Automatically generated (only) if no simulation parameters file is input.* |
| `Simulation_data.csv` | Contains the simulated impedance values at each frequency, including Re(Z), Im(Z), −Im(Z), \|Z\|, phase, and classic export columns. |
| `Simulation_nyquist.png` | Nyquist plot of the simulated impedance spectrum, showing Re(Z) versus −Im(Z). |
| `Simulation_bode_magnitude.png` | Bode magnitude plot of the simulated impedance spectrum, showing simulated \|Z\| versus frequency. |
| `Simulation_bode_phase.png` | Bode phase plot of the simulated impedance spectrum, showing simulated phase angle versus frequency. |

### Simulation procedure

1. Select **Simulation** (Files and circuit tab).
2. Select an output folder.
3. Enter the equivalent circuit (see Section 4).
4. Run program to generate Simulation parameters template file (optional)
5. Select Simulation parameters CSV file (Simulation tab).
6. Click RUN SIMULATION.

## 4.3 Data Viewer

Loads an experimental impedance data file, calculates derived quantities such as |Z| and phase, and displays/saves Nyquist and Bode plots without fitting a model.

**Input:**

- `Data file` (.txt)
- `Output folder`: Folder where all the output files will be saved

**Output:**

| File | Description |
|---|---|
| `View_Data.csv` | Contains the loaded experimental impedance spectrum data with derived quantities such as Re(Z), Im(Z), −Im(Z), \|Z\|, phase, and classic export columns. |
| `View_Data_Nyquist.png` | Nyquist plot of the experimental data, showing Re(Z) versus −Im(Z) without any model fit. |
| `View_Data_Bode_Magnitude.png` | Bode magnitude plot of the experimental data, showing measured \|Z\| versus frequency. |
| `View_Data_Bode_Phase.png` | Bode phase plot of the experimental data, showing measured phase angle versus frequency. |

### View Data procedure

7. Select **Data Viewer** (Files and circuit tab).
8. Select data file.
9. Select an output folder.
10. Click VIEW DATA.

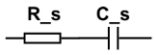
## 4. Equivalent-circuit syntax

| Operator | Meaning | Example |
|---|---|---|
| + | Series connection | `R_s + R_1` |
| \|\| | Parallel connection | `R_1 \|\| CPE_1` |
| (...) | Grouping | `R_s + (R_1\|\|CPE_1)` |

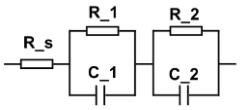
Spaces do not matter. Element names must be written as **`TYPE_label`**. Bare names such as `Rs`, `R1`, `CPE1`, and `WsIon` are intentionally rejected. Valid examples: `R_s`, `R_1`, `C_1`, `CPE_1`, `Ws_ion`, `U_custom`

**Example Circuits:**

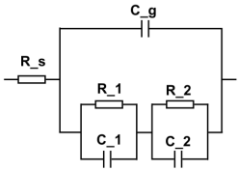
`R_s + C_s`




`R_s + (R_1||C_1) + (R2_C2)`




`R_s + (C_g||((R_1||C_1) + (R_1||C_1)))`




### Supported elements and parameters

| Element TYPE | Meaning | Parameters (unit) |
|---|---|---|
| R | Resistor | R (Ω) |
| C | Ideal capacitor | C (F) |
| L | Ideal inductor | L (H) |
| CPE | Constant phase element | Q ($S \cdot s^n$), n (-) |
| LPE | Inductive phase element | QL_label, nL_label |
| W | Semi-infinite Warburg | W ($\Omega/s^{1/2}$) |
| Ws | Finite-length short/transmissive Warburg | Wsr ($\Omega/s^{1/2}$), Wsc ($s^{1/2}$) |
| Wo | Finite-length open/reflective Warburg | Wor ($\Omega/s^{1/2}$), Woc ($s^{1/2}$) |
| G | Gerischer element | Yg ($S \cdot s^{-1/2}$), Kg ($s^{-1}$) |
| U | User-defined element | Default: UP1_label, UP2_label, …. |

| Element TYPE | Mathematical convention |
|---|---|
| R | Z= R |
| C | Z = 1/(j*ω*C) |
| L | Z = j* ω *L |
| CPE | Z = Q^-1*(j* ω)^(-n) |
| LPE | Z = QL*(j* ω)^(nL) |
| W | Z = W*(1-j)/sqrt(ω) |
| Ws | Z = Wsr*(1-j)/sqrt(ω) * tanh(Wsc*sqrt(j* ω)) |
| Wo | Z = Wor*(1-j)/sqrt(ω) * coth(Woc*sqrt(j* ω)) |
| G | Z = 1/(Yg*sqrt(Kg + j* ω)) |
| U | |

Note: In this convention Wsr/Wor are Warburg coefficients and Wsc/Woc $= \delta/\sqrt{D}$, where $\delta$ is the diffusion length and *D* is the diffusion coefficient.

For the User Defined element see Section 5.

## 5. User-defined element tab

The User element tab allows custom impedance elements to be defined without editing the Python source code. The program supports up to three GUI-defined user elements. Use strict names such as U_1, U_2, U_fast, U_ion, or U_custom. Do not use U1 or Ufast.

### Expression variables

For each user element you enter the element name, number of parameters, Re(Z) expression, and Im(Z) expression. The imaginary expression is Im(Z), not -Im(Z).

```
Available variables:
f       ordinary frequency in Hz
omega   angular frequency in rad/s
w       same as omega
P1...P5 local user-element parameters

Available functions include:
sqrt, exp, log, log10, sin, cos, tan, sinh, cosh, tanh, abs, minimum, maximum, where
```

### Parameter naming

| Circuit element | Local variables | CSV parameter names |
|---|---|---|
| U_1 | P1, P2 | P1_1, P2_1 |
| U_2 | P1, P2, P3 | P1_2, P2_2, P3_2 |
| U_ion | P1, P2 | P1_ion, P2_ion |
| U_trap | P1 | P1_trap |

### Examples

#### Pure resistor user element:

```
Element name: U_res
Number of parameters: 1
Re(Z): P1
Im(Z): 0
Circuit: R_s + U_res
```

**Capacitor-like user element:**

```
Element name: U_cap
Number of parameters: 1
Re(Z): 0
Im(Z): -1/(omega*P1)
Circuit: R_s + U_cap
```

**Two user elements in one circuit:**

```
U_1:
  Number of parameters: 2
  Re(Z): P1
  Im(Z): -1/(omega*P2)

U_2:
  Number of parameters: 1
  Re(Z): P1
  Im(Z): 0

Circuit: R_s + U_1 + U_2
```

### Practical cautions

- Use physically motivated formulas.
- Test user elements first in Simulation mode.
- Use the fewest parameters needed.
- Apply sensible bounds in fit mode.
- Check stability against random seed and robust loss settings.
- Do not use multiple high-parameter U elements unless there is a clear physical reason.

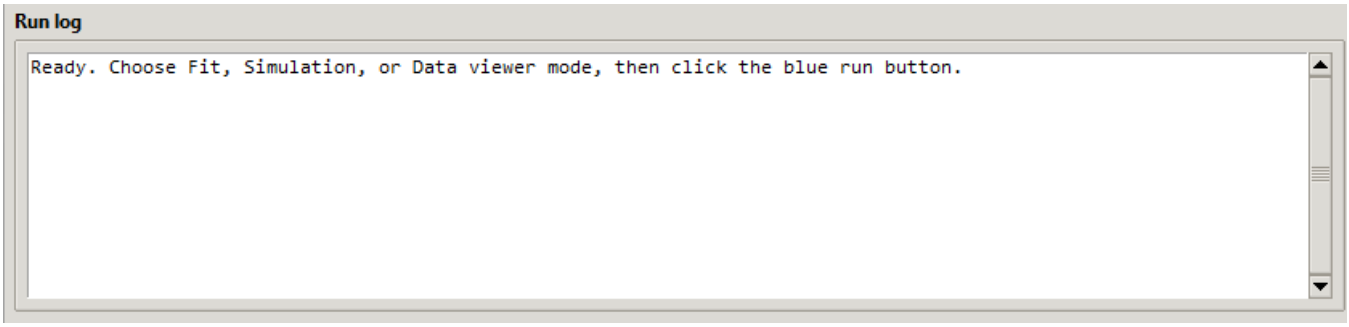
## 6. Output Options

The Output options control how the program saves and displays the results generated by each mode. They let the user decide whether CSV/PNG files should be written to the selected output folder, set the resolution of saved plot images through the DPI value, and optionally open the saved figures automatically after a run. These settings do not change the fitting, simulation, or data-viewer calculations themselves; they only control how the results are exported and presented.

- `Save output files`: When checked, the program writes the generated CSV files and PNG plots to the selected output folder; when unchecked, the run can be used mainly for viewing results in the GUI without saving new output files.
- `Plot DPI`: Sets the resolution of the saved PNG plots; higher values give sharper figures for reports/papers but larger file sizes, while 300 DPI is a good default for publication-quality images.
- `Open saved plot images after run`: When checked, the program automatically opens the saved PNG plots in the operating-system image viewer after the run finishes; the plots are still shown inside the GUI either way.
- `Open output folder` button is present on the bottom right of the GUI window. Click it to open the currently selected output folder.

## 7. Run Log

The Run log window shows the real-time text output generated while the program is running. It reports the selected operation mode, loaded files, parsed circuit elements, active restriction files, fitting progress, warnings, errors, generated output files, fit metrics, and final parameter values. It is mainly used for checking that the run used the intended settings and for diagnosing problems if the fit, simulation, or data-viewer operation fails.




## 8. Recommended practical workflow

1. Use Data viewer mode to check that the file loads correctly and the raw plots look sensible.
2. Run Fit mode once without restriction CSVs to generate templates.
3. Edit parameter_restrictions_template.csv to set physically reasonable values and bounds.
4. Optionally edit derived_restrictions_template.csv to constrain branch frequencies or Warburg scales.
5. Fit again using the edited CSV files.
6. Inspect Nyquist/Bode overlays, residuals, metrics, and whether parameters hit bounds.
7. Rerun with different random seeds and with huber loss as a robustness check.
8. Use Simulation mode to verify how parameter changes affect the expected spectrum.

## 9. Troubleshooting

| Problem | Likely cause | Fix |
|---|---|---|
| Circuit parsing error | Element names do not follow TYPE_label. | Use names like R_1, CPE_1, Ws_ion. |
| No numeric rows found | File format is not Re(Z), -Im(Z), Frequency. | Check columns, delimiters, and header lines. |
| Fit gives crazy values | Bounds too loose or branch swapping. | Use parameter and derived restriction CSVs. |
| Fit changes strongly with random seed | Multiple local minima or non-identifiability. | Tighten bounds, add derived restrictions, increase random starts. |
| Parameters hit bounds | Bounds may be too tight or model is forcing a limit. | Check whether the bound is intentional. |
| Simulation only writes a template | No simulation parameter CSV was selected. | Edit and select simulation_parameters_template.csv. |
| Low-frequency region fits poorly | Drift, non-stationarity, missing physics, or weakly identifiable Warburg branch. | Compare robust losses, inspect residuals, consider a frequency window or modified circuit. |